

ASSIGNMENT-7.5

HT.NO:2303A51732

Task 1 (Mutable Default Argument – Function Bug)

Task: Analyze given code where a mutable default argument causes unexpected behavior. Use AI to fix it.

Bug: Mutable default argument

```
def add_item(item, items=[]):
```

```
    items.append(item)
```

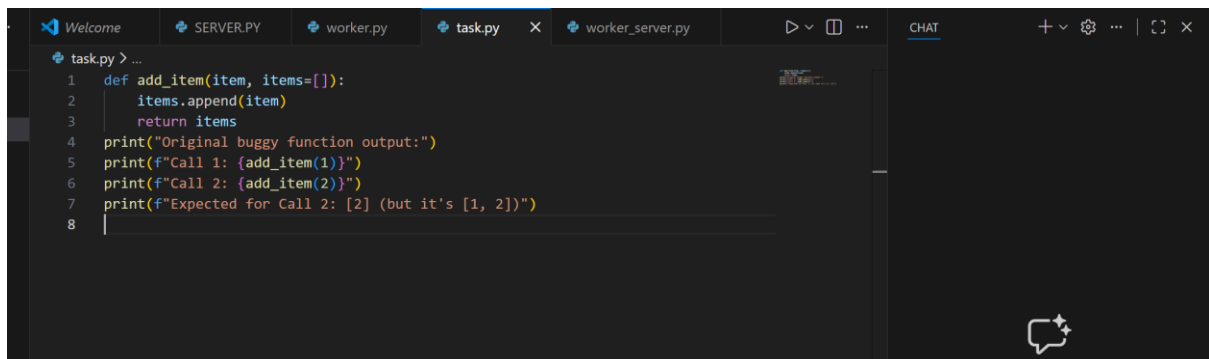
```
    return items
```

```
print(add_item(1))
```

```
print(add_item(2))
```

Expected Output: Corrected function avoids shared list bug

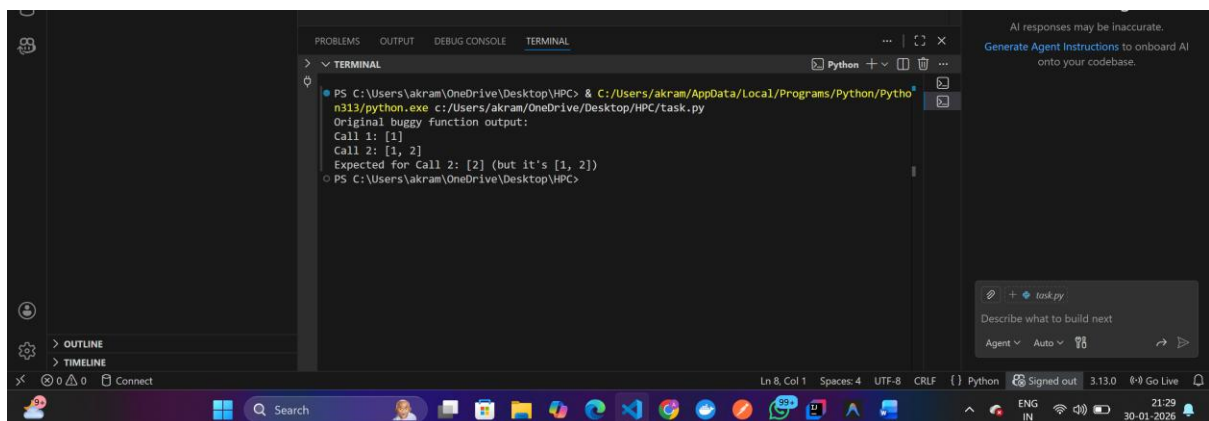
Original Buggy Code:

A screenshot of a code editor window with a dark theme. The editor shows a file named 'task.py' with the following Python code:

```
1 def add_item(item, items=[]):
2     items.append(item)
3     return items
4
5 print("Original buggy function output:")
6 print(f"Call 1: {add_item(1)}")
7 print(f"Call 2: {add_item(2)}")
8 print(f"Expected for Call 2: [2] (but it's [1, 2])")
```

The editor has tabs for 'Welcome', 'SERVER.PY', 'worker.py', 'task.py', and 'worker_server.py'. The 'task.py' tab is active. On the right side, there is a 'CHAT' panel with a plus icon and a star icon.

Output:

A screenshot of a terminal window in a code editor. The terminal shows the output of running the Python code from the previous block. The command executed is 'python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py'. The output is:

```
Original buggy function output:
Call 1: [1]
Call 2: [1, 2]
Expected for Call 2: [2] (but it's [1, 2])
```

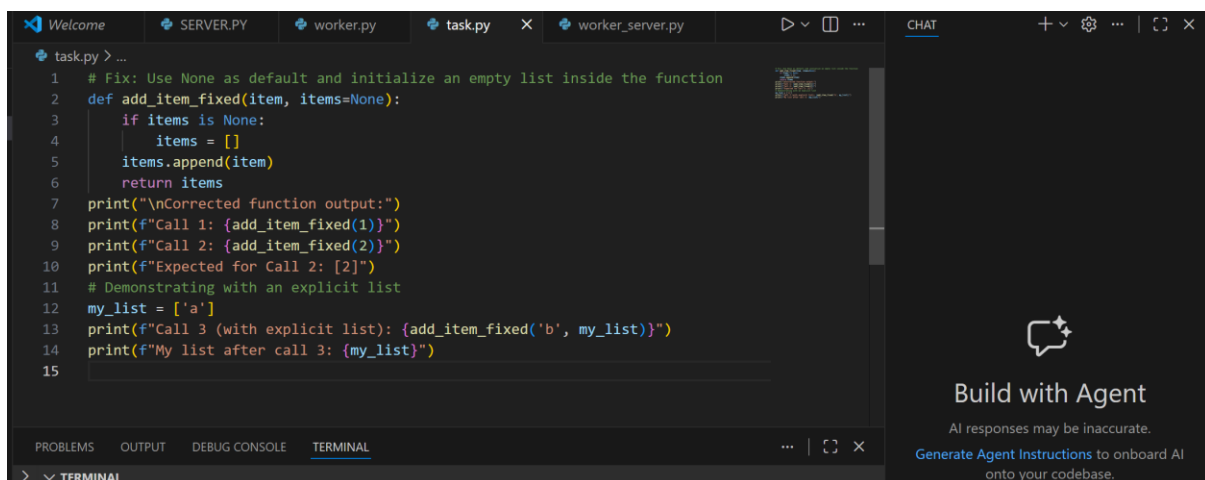
The terminal window has tabs for 'PROBLEMS', 'OUTPUT', 'DEBUG CONSOLE', and 'TERMINAL'. The 'TERMINAL' tab is active. On the right side, there is a panel with the text 'AI responses may be inaccurate. Generate Agent Instructions to onboard AI onto your codebase.' and a button to '+ task.py'. Below this, there is a section 'Describe what to build next' with 'Agent' and 'Auto' options.

Explanation of the Bug:

The `add_item` function uses a list `[]` as a default argument for `items`. In Python, default arguments are evaluated only once, when the function is defined, not every time the function is called. This means that all calls to `add_item` without explicitly providing an `items` list will share the *same* list object.

When `add_item(1)` is called, 1 is appended to the shared default list. When `add_item(2)` is called *again without a new list*, it appends 2 to the *same* list that already contains 1, resulting in `[1, 2]` instead of the expected `[2]` (if each call was meant to start with an empty list).

Corrected Code:



```
1 # Fix: Use None as default and initialize an empty list inside the function
2 def add_item_fixed(item, items=None):
3     if items is None:
4         items = []
5         items.append(item)
6     return items
7
8 print("\nCorrected function output:")
9 print(f"Call 1: {add_item_fixed(1)}")
10 print(f"Call 2: {add_item_fixed(2)}")
11 print(f"Expected for Call 2: [2]")
12
13 # Demonstrating with an explicit list
14 my_list = ['a']
15 print(f"Call 3 (with explicit list): {add_item_fixed('b', my_list)}")
16 print(f"My list after call 3: {my_list}")
```

Output:



```
Call 1: [1]
Call 2: [1, 2]
Expected for Call 2: [2] (but it's [1, 2])
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py

Corrected function output:
Call 1: [1]
Call 2: [2]
Expected for Call 2: [2]
Call 3 (with explicit list): ['a', 'b']
My list after call 3: ['a', 'b']
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Fix:

The common and recommended fix for mutable default arguments is to use `None` as the default value. Inside the function, check if `items` is `None`, and if it is, then initialize an empty list `items = []`. This ensures that a new, empty list is created each time the function is called without an explicit `items` argument, preventing the shared list issue.

Task 2 (Floating-Point Precision Error)

Task: Analyze given code where floating-point comparison fails.

Use AI to correct with tolerance.

```
2 def check_sum():
3     return (0.1 + 0.2) == 0.3
4     print(check_sum())
5     Expected Output: Corrected function
6     Original Buggy Code:
7     # Bug: Floating point precision issue
8     def check_sum():
9         return (0.1 + 0.2) == 0.3
10    print("Original buggy function output:")
11    print(f"Result: {check_sum()}")
12    print(f"Expected: True")
13
```

Output:

```
Call 2: [2]
Expected for Call 2: [2]
Call 3 (with explicit list): ['a', 'b']
My list after call 3: ['a', 'b']
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:\Users\akram\AppData\Local\Programs\Python\Python310\python.exe c:\Users\akram\OneDrive\Desktop\HPC\task.py
File "c:\Users\akram\OneDrive\Desktop\HPC\task.py", line 3
    return (0.1 + 0.2) == 0.3
    ^^^^^^
IndentationError: expected an indented block after function definition on line 2
```

Explanation of the Bug:

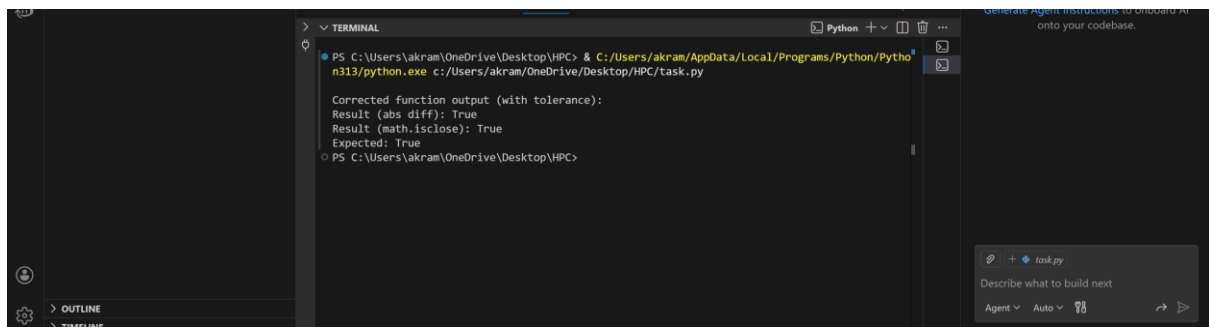
Floating-point numbers (like 0.1, 0.2, 0.3) are represented in computers using binary fractions. Most decimal fractions, such as 0.1, cannot be represented exactly in binary. This leads to tiny precision errors.

When $0.1 + 0.2$ is computed, the result is not exactly 0.3, but a number very close to it (e.g., 0.30000000000000004). Directly comparing this slightly imprecise sum with the exactly represented 0.3 using `==` will return False, even though mathematically they should be equal.

Corrected Code:

```
3 def check_sum_fixed():
7     expected_val = 0.3
8     # Compare if the absolute difference is within the tolerance
9     return abs(sum_val - expected_val) < tolerance
10 # Alternative using math.isclose() (Python 3.5+)
11 def check_sum_isclose():
12     return math.isclose(0.1 + 0.2, 0.3)
13 print("\nCorrected function output (with tolerance):")
14 print(f"Result (abs diff): {check_sum_fixed()}")
15 print(f"Result (math.isclose): {check_sum_isclose()}")
16 print(f"Expected: True")
17
```

Output:



```
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py

Corrected function output (with tolerance):
Result (abs diff): True
Result (math.isclose): True
Expected: True
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

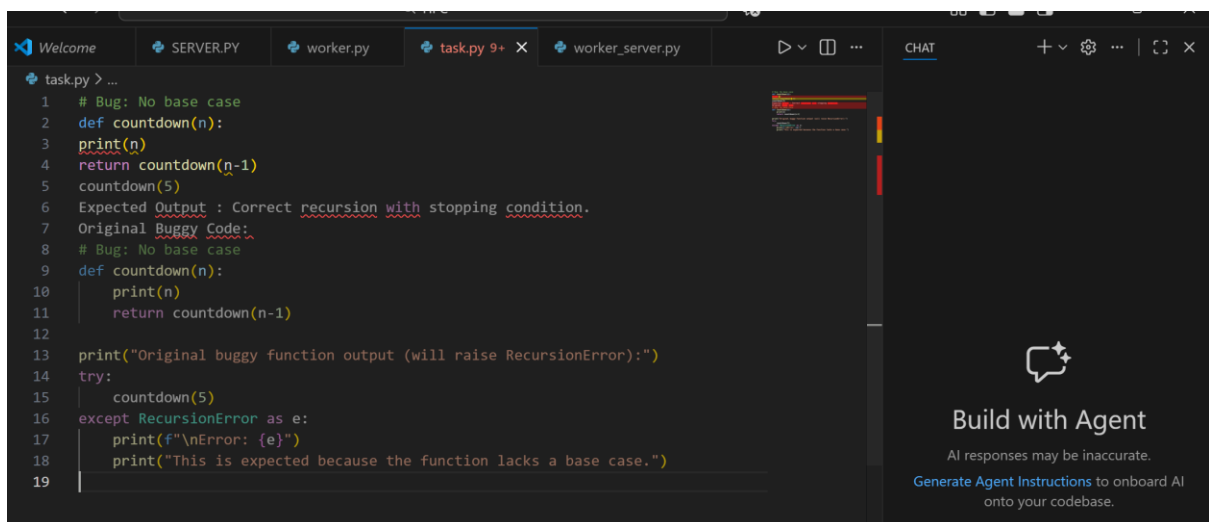
Explanation of the Fix:

To correctly compare floating-point numbers, instead of checking for exact equality, we check if their absolute difference is less than a small tolerance value (often called epsilon). If the difference is smaller than this tolerance, the numbers are considered practically equal.

Python's math module also provides `math.isclose()`, which is a convenient and robust way to perform such comparisons, taking into account both relative and absolute tolerances.

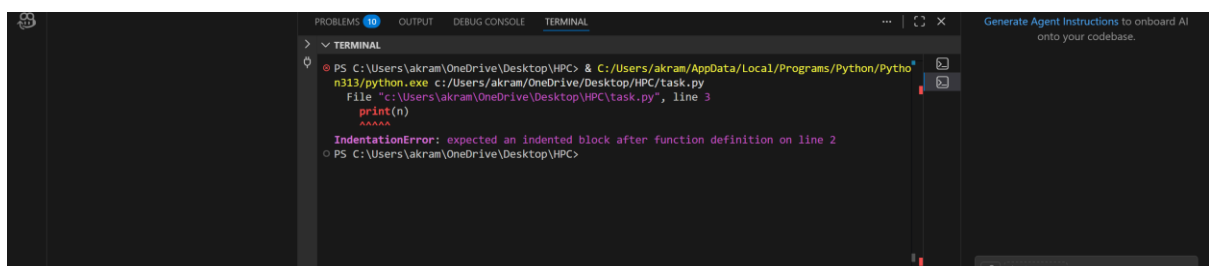
Task 3 (Recursion Error – Missing Base Case)

Task: Analyze given code where recursion runs infinitely due to missing base case. Use AI to fix.



```
task.py > ...
1 # Bug: No base case
2 def countdown(n):
3     print(n)
4     return countdown(n-1)
5     countdown(5)
6 Expected Output : Correct recursion with stopping condition.
7 Original Buggy Code:
8 # Bug: No base case
9 def countdown(n):
10     print(n)
11     return countdown(n-1)
12
13 print("Original buggy function output (will raise RecursionError):")
14 try:
15     countdown(5)
16 except RecursionError as e:
17     print(f"\nError: {e}")
18     print("This is expected because the function lacks a base case.")
19
```

Output:



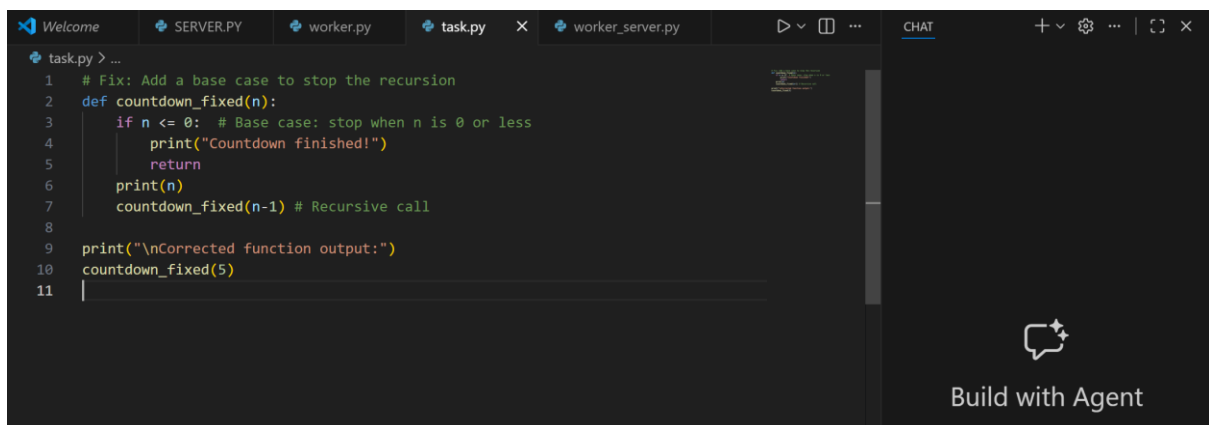
```
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
File "c:\Users\akram\OneDrive\Desktop\HPC\task.py", line 3
    print(n)
    ^^^^^
IndentationError: expected an indented block after function definition on line 2
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Bug:

Recursion is a programming technique where a function calls itself to solve a problem. Every recursive function *must* have a **base case**, which is a condition that stops the recursion. Without a base case, the function will call itself infinitely, leading to an ever-growing stack of function calls.

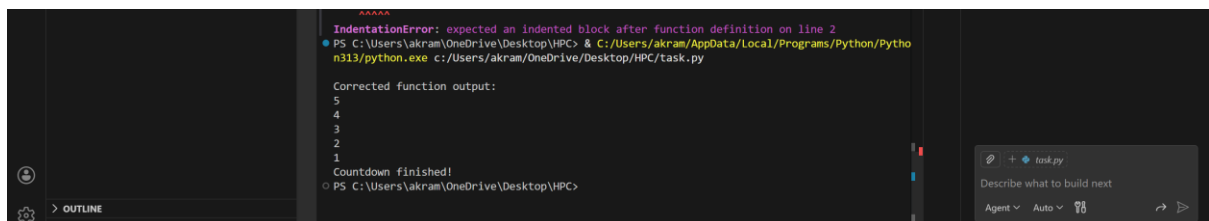
In the countdown function, there's no condition to stop the countdown(n-1) call. When n reaches 0, it continues to call countdown(-1), countdown(-2), and so on. Eventually, Python's call stack limit is exceeded, resulting in a RecursionError.

Corrected Code:



```
task.py > ...
1 # Fix: Add a base case to stop the recursion
2 def countdown_fixed(n):
3     if n <= 0: # Base case: stop when n is 0 or less
4         print("Countdown finished!")
5         return
6     print(n)
7     countdown_fixed(n-1) # Recursive call
8
9 print("\nCorrected function output:")
10 countdown_fixed(5)
11 |
```

Output:



```
*****
IndentationError: expected an indented block after function definition on line 2
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py

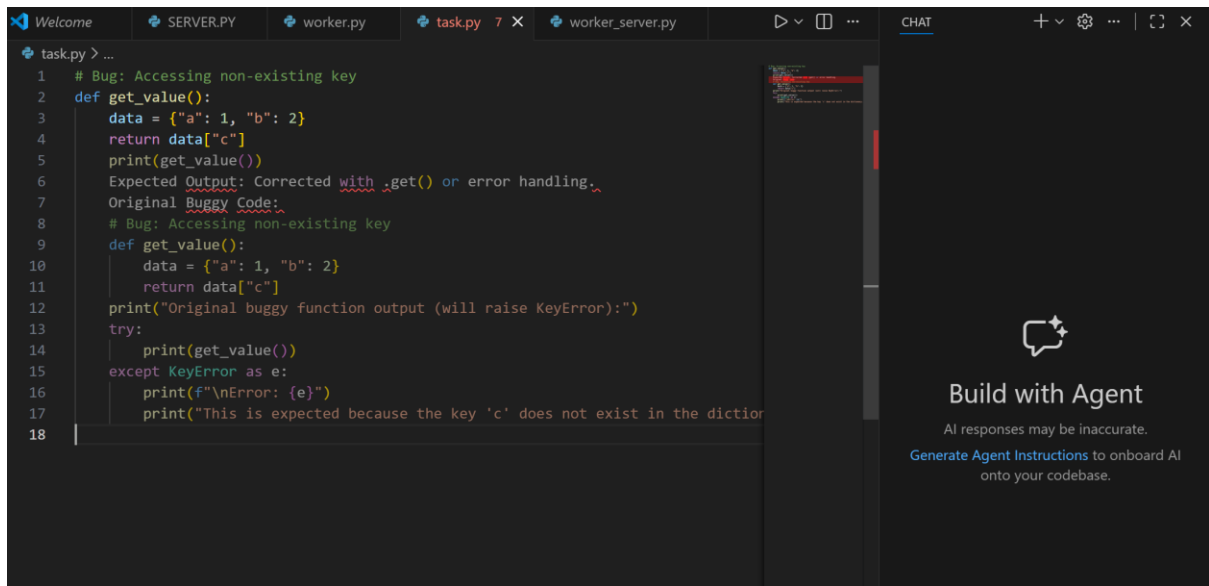
Corrected function output:
5
4
3
2
1
Countdown finished!
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Fix:

The fix involves adding an if n <= 0: condition at the beginning of the countdown_fixed function. This is our base case. When n becomes 0 or less, the function prints "Countdown finished!" and then returns, effectively stopping the chain of recursive calls. This prevents the RecursionError and ensures the function behaves as intended.

Task 4 (Dictionary Key Error)

Task: Analyze given code where a missing dictionary key causes error. Use AI to fix it.



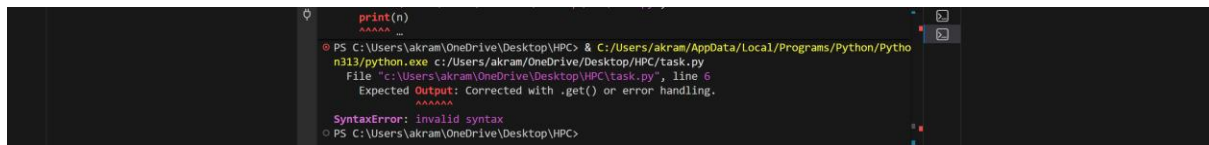
```
task.py > ...
1 # Bug: Accessing non-existing key
2 def get_value():
3     data = {"a": 1, "b": 2}
4     return data["c"]
5     print(get_value())
6     Expected Output: Corrected with .get() or error handling.
7     Original Buggy Code:
8     # Bug: Accessing non-existing key
9     def get_value():
10        data = {"a": 1, "b": 2}
11        return data["c"]
12        print("Original buggy function output (will raise KeyError):")
13        try:
14            print(get_value())
15        except KeyError as e:
16            print(f"\nError: {e}")
17        print("This is expected because the key 'c' does not exist in the dictionary")
18
```

Build with Agent

AI responses may be inaccurate.

[Generate Agent Instructions](#) to onboard AI onto your codebase.

Output:



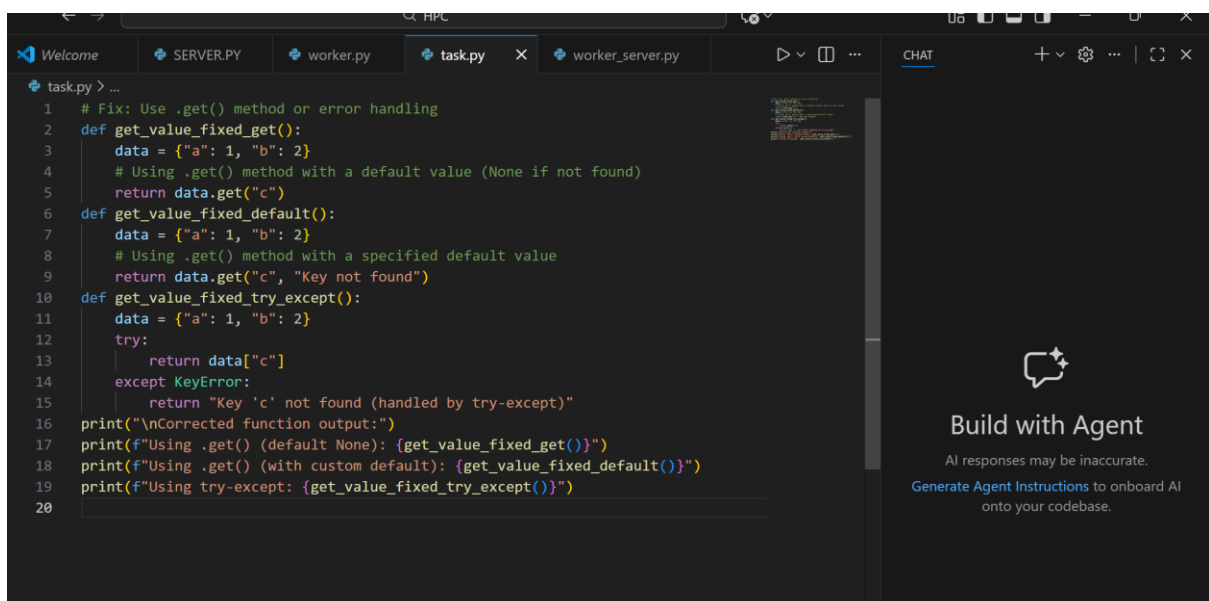
```
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
File "c:\Users\akram\OneDrive\Desktop\HPC\task.py", line 6
Expected Output: Corrected with .get() or error handling.
SyntaxError: invalid syntax
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Bug:

A `KeyError` occurs when you try to access a dictionary using a key that does not exist within that dictionary. In Python, attempting `dictionary[key]` when `key` is not present will immediately raise a `KeyError`, interrupting the program's execution.

In the `get_value` function, the dictionary `data` contains keys `"a"` and `"b"`, but there is no key `"c"`. Therefore, `data["c"]` raises a `KeyError`.

Corrected Code:



```
task.py > ...
1 # Fix: Use .get() method or error handling
2 def get_value_fixed_get():
3     data = {"a": 1, "b": 2}
4     # Using .get() method with a default value (None if not found)
5     return data.get("c")
6 def get_value_fixed_default():
7     data = {"a": 1, "b": 2}
8     # Using .get() method with a specified default value
9     return data.get("c", "Key not found")
10 def get_value_fixed_try_except():
11     data = {"a": 1, "b": 2}
12     try:
13         return data["c"]
14     except KeyError:
15         return "Key 'c' not found (handled by try-except)"
16 print("\nCorrected function output:")
17 print(f"Using .get() (default None): {get_value_fixed_get()}")
18 print(f"Using .get() (with custom default): {get_value_fixed_default()}")
19 print(f"Using try-except: {get_value_fixed_try_except()}")
20
```

Build with Agent

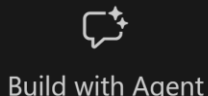
AI responses may be inaccurate.

[Generate Agent Instructions](#) to onboard AI onto your codebase.

Output:

```
n313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py

Corrected function output:
Using .get() (default None): None
Using .get() (with custom default): Key not found
Using try-except: Key 'c' not found (handled by try-except)
PS C:\Users\akram\OneDrive\Desktop\HPC>
```



Explanation of the Fix:

There are two common ways to handle missing dictionary keys gracefully:

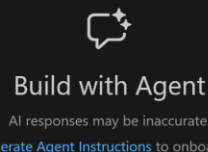
1. **Using the .get() method:** Instead of `dictionary[key]`, you can use `dictionary.get(key)`. If key exists, it returns its corresponding value. If key does not exist, it returns `None` by default, or a specified default value if provided (e.g., `dictionary.get(key, 'default_value')`). This avoids raising a `KeyError`.
2. **Using a try-except block:** You can wrap the dictionary access `dictionary[key]` within a try block. If a `KeyError` occurs, it will be caught by the except `KeyError` block, where you can define how to handle the error (e.g., return a default value, log the error, or raise a different exception).

Task 5 (Infinite Loop – Wrong Condition)

Task: Analyze given code where loop never ends. Use AI to detect

and fix it.

```
task.py > ...
1 # Bug: Infinite loop
2 def loop_example():
3     i = 0
4     while i < 5:
5         print(i)
6 def loop_example():
7     i = 0
8     while i < 5:
9         print(i)
10
11 print("Original buggy function output (will run infinitely if not stopped):")
12
```



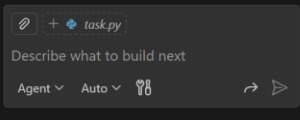
Calling the function to demonstrate the infinite loop.

You might need to manually stop the execution of this cell if it runs too long.

`loop_example()`

Output:

```
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
File "c:\Users\akram\OneDrive\Desktop\HPC\task.py", line 5
    print(i)
    ^^^^^
IndentationError: expected an indented block after 'while' statement on line 4
PS C:\Users\akram\OneDrive\Desktop\HPC>
```



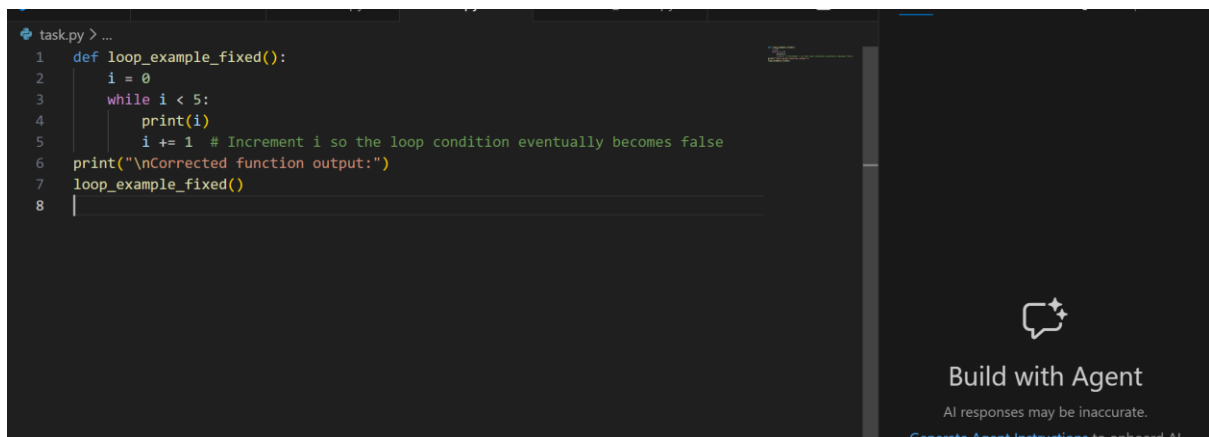
Explanation of the Bug:

An infinite loop occurs when the condition that controls the loop never becomes false. In this while loop, the variable `i` is initialized to 0, and the loop continues as long as `i < 5`. However, there is no code inside the loop that modifies the value of `i`. As a result, `i` remains 0 indefinitely, and the condition `i < 5` is always true, causing the loop to print 0 repeatedly without ever terminating.

Corrected Code:

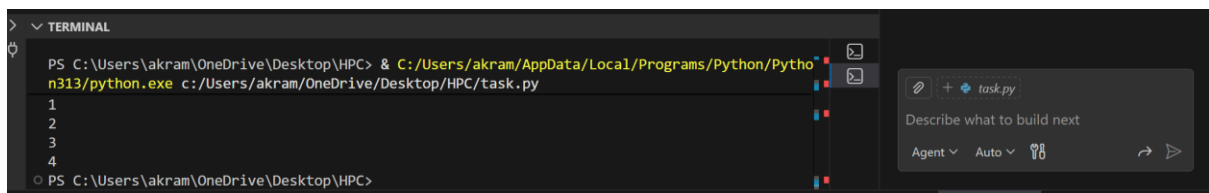
Fix: Increment the loop counter within the loop

```
task.py > ...
1 def loop_example_fixed():
2     i = 0
3     while i < 5:
4         print(i)
5         i += 1 # Increment i so the loop condition eventually becomes false
6 print("\nCorrected function output:")
7 loop_example_fixed()
8 |
```



Output:

```
> TERMINAL
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
1
2
3
4
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

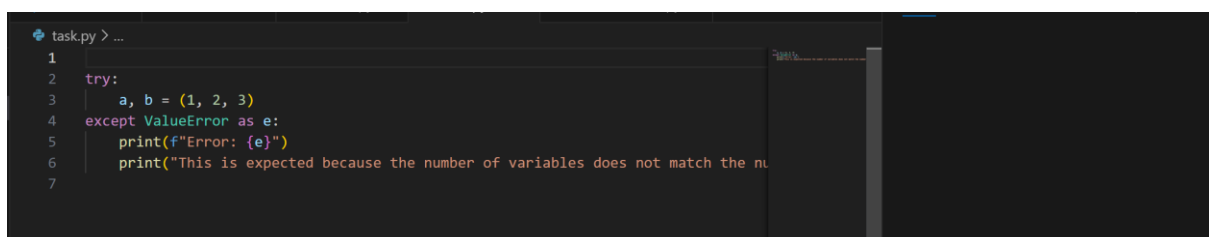


Explanation of the Fix:

The fix involves adding `i += 1` inside the while loop. This statement increments the value of `i` in each iteration. With `i` increasing, it will eventually reach 5 (or greater), causing the loop condition `i < 5` to become false, and the loop will terminate as intended. This ensures that the loop executes a finite number of times.

Task 6 (Unpacking Error – Wrong Variables)

```
task.py > ...
1
2 try:
3     a, b = (1, 2, 3)
4 except ValueError as e:
5     print(f"Error: {e}")
6     print("This is expected because the number of variables does not match the number of values")
7
```



Output:

```
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
Error: too many values to unpack (expected 2)
This is expected because the number of variables does not match the number of values.
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Bug:

In Python, sequence unpacking allows you to assign elements of an iterable (like a tuple or list) to multiple variables simultaneously. A `ValueError` occurs during unpacking when the number of variables on the left-hand side of the assignment does not match the number of elements in the sequence on the right-hand side.

In this buggy code, we attempt to unpack a tuple `(1, 2, 3)` (which has three elements) into only two variables, `a` and `b`. This mismatch causes Python to raise a `ValueError`, indicating that there are too many values to unpack.

Corrected Code:

```
task.py > ...
1 # Fix: Match the number of variables to the number of values, or use extended unpacking
2 # Option 1: Match the number of variables
3 x, y, z = (1, 2, 3)
4 print(f"Correct unpacking (match variables): x={x}, y={y}, z={z}")
5 # Option 2: Use underscore '_' for unwanted values (if you only need a subset)
6 a, b, _ = (1, 2, 3) # Unpacks 1 into a, 2 into b, and 3 into the discard variable
7 print(f"Correct unpacking (discarding with _): a={a}, b={b}")
8 # Option 3: Use extended unpacking (Python 3+) for flexible assignments
9 # If you only need the first two and want to catch the rest
10 first, second, *rest = (1, 2, 3, 4, 5)
11 print(f"Correct unpacking (extended unpacking, *rest): first={first}, second={second}, rest={rest}")
12 # If you only need the first two and explicitly discard the rest
13 value1, value2, *_ = (10, 20, 30, 40)
14 print(f"Correct unpacking (extended unpacking, *_): value1={value1}, value2={value2}")
15
```

Output:

```
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:/Users/akram/AppData/Local/Programs/Python/Python313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
Correct unpacking (match variables): x=1, y=2, z=3
Correct unpacking (discarding with _): a=1, b=2
Correct unpacking (extended unpacking, *rest): first=1, second=2, rest=[3, 4, 5]
Correct unpacking (extended unpacking, *_): value1=10, value2=20
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Fix:

There are several ways to fix an unpacking error, depending on your intent:

1. **Match the number of variables:** The most straightforward fix is to ensure that the number of variables on the left-hand side exactly matches the number of elements in the sequence being unpacked. If the sequence has three elements, you need three variables.
2. **Use `_` for unwanted values:** If you only care about a subset of the values in the sequence, you can use the underscore `_` as a placeholder variable for the elements

you want to ignore. This is a convention in Python to indicate a variable whose value is not going to be used.

3. **Use extended unpacking (* operator):** For more flexible unpacking, especially with sequences of unknown length or when you want to capture multiple remaining items, Python 3+ allows the use of the * operator (e.g., *rest). This will collect all remaining items into a list. You can also use *_ to discard multiple remaining items explicitly.

Original Buggy Code:

```
task.py > ...
1 # Bug: Mixed indentation
2 def func():
3     x = 5
4     y = 10 # This line has a tab for indentation
5     return x+y
6 print("Original buggy function output (will raise IndentationError):")
7 try:
8     func()
9 except IndentationError as e:
10    print(f"\nError: {e}")
11    print("This is expected due to inconsistent use of tabs and spaces for indent
12
```

Output:

```
IndentationError: unindent does not match any outer indentation level
PS C:\Users\akram\OneDrive\Desktop\HPC> & C:\Users\akram\AppData\Local\Programs\Python\Python
n313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
File "c:\Users\akram\OneDrive\Desktop\HPC\task.py", line 4
    y = 10 # This line has a tab for indentation
    ^
IndentationError: unindent does not match any outer indentation level
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Bug:

Python uses indentation to define code blocks. Mixing tabs and spaces for indentation within the same file or even within the same code block is a common source of IndentationError. While Python 3 largely warns against and often prohibits such mixing, older Python 2 versions might interpret them differently, leading to subtle logic errors. Even in Python 3, if a tab and spaces are used to define what should be the same indentation level, Python raises an IndentationError because it cannot consistently determine the block structure.

In the func function, the line `x = 5` is indented with spaces, while the line `y = 10` is intended to be at the same level but is indented with a tab. This inconsistency causes Python to raise an IndentationError.

Correceted Code:

Fix: Ensure consistent indentation (use only spaces or only tabs, preferably spaces)

```
task.py > ...
1 def func_fixed():
2     x = 5
3     y = 10 # Corrected to use spaces consistently
4     return x+y
5 print("\nCorrected function output:")
6 print(f"Result: {func_fixed()}")
7
```

Output:

```
n313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
Corrected function output:
Result: 15
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Fix:

The fix involves ensuring consistent indentation throughout the code. The Python community standard (PEP 8) recommends using 4 spaces per indentation level. By replacing the tab with spaces (or vice-versa, as long as it's consistent), the `IndentationError` is resolved, and the code runs as expected.

Task 8 (Import Error – Wrong Module Usage)

Task: Analyze given code with incorrect import. Use AI to fix.

```
Welcome SERVER.PY worker.py task.py 1 x worker_server.py
task.py > ...
1 # Bug: Wrong import
2 import maths
3 print(maths.sqrt(16))
4
5 print("Original buggy function output (will raise ModuleNotFoundError):")
6 try:
7     print(maths.sqrt(16))
8 except ModuleNotFoundError as e:
9     print(f"\nError: {e}")
10    print("This is expected because the module 'maths' does not exist (typo for 'math')")
11
```

Output:

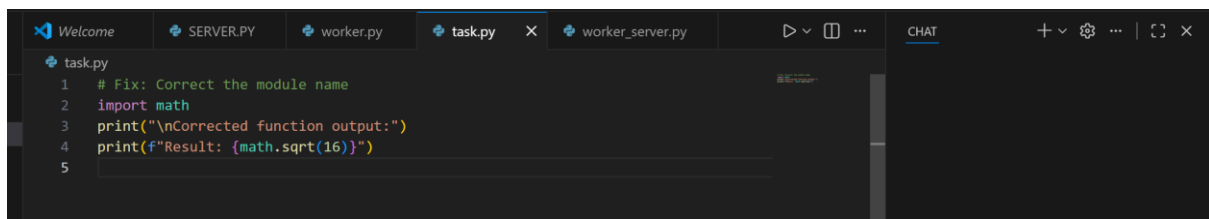
```
PS C:\Users\akram\OneDrive\Desktop\HPC> n313/python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
Traceback (most recent call last):
  File "c:/Users/akram/OneDrive/Desktop/HPC/task.py", line 2, in <module>
    import maths
ModuleNotFoundError: No module named 'maths'
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Bug:

A `ModuleNotFoundError` occurs when Python cannot find the module you are trying to import. This can happen due to several reasons, including a typo in the module name, the module not being installed, or issues with Python's path.

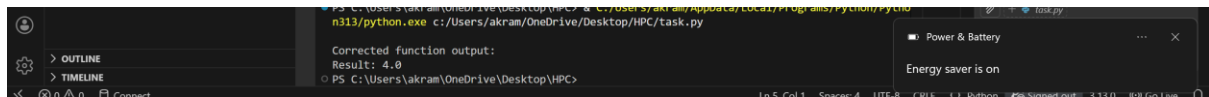
In this specific buggy code, the error is a simple typo: `maths` instead of the correct module name `math`. When Python attempts to import `maths`, it cannot locate a module with that name and thus raises a `ModuleNotFoundError`.

Corrected Code:



```
task.py
1 # Fix: Correct the module name
2 import math
3 print("\nCorrected function output:")
4 print(f"Result: {math.sqrt(16)}")
5
```

Output:



```
PS C:\Users\akram\OneDrive\Desktop\HPC> C:\Users\akram\AppData\Local\Programs\Python\Python313\python.exe c:/Users/akram/OneDrive/Desktop/HPC/task.py
Corrected function output:
Result: 4.0
PS C:\Users\akram\OneDrive\Desktop\HPC>
```

Explanation of the Fix:

The fix is straightforward: correct the typo in the import statement from `import maths` to `import math`. The `math` module is a standard Python library that provides mathematical functions, including `sqrt` for square root. Once the correct module is imported, its functions can be called without error.